The AI Scientist-v2: Workshop-Level Automated Scientific Discovery via Agentic Tree Search

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AI is increasingly playing a pivotal role in transforming how scientific discoveries are made. We introduce The AI Scientist-v2, an end-to-end agentic system capable of producing the first entirely AI-generated peer-review-accepted workshop paper. This system iteratively formulates scientific hypotheses, designs and executes experiments, analyzes and visualizes data, and autonomously authors scientific manuscripts. Compared to its predecessor (v1, Lu et al., 2024), The AI Scientist-v2 eliminates the reliance on human-authored code templates, generalizes effectively across diverse machine learning domains, and leverages a novel progressive agentic tree-search methodology managed by a dedicated experiment manager agent. Additionally, we enhance the AI reviewer component by integrating a Vision-Language Model (VLM) feedback loop for iterative refinement of content and aesthetics of the figures. We evaluated The AI Scientist-v2 by submitting three fully autonomous manuscripts to a peer-reviewed ICLR workshop. Notably, one manuscript achieved high enough scores to exceed the average human acceptance threshold, marking the first instance of a fully AI-generated paper successfully navigating a peer review. This accomplishment highlights the growing capability of AI in conducting all aspects of scientific research. We anticipate that further advancements in autonomous scientific discovery technologies will profoundly impact human knowledge generation, enabling unprecedented scalability in research productivity and significantly accelerating scientific breakthroughs, greatly benefiting society at large. We have open-sourced the code at https://github.com/SakanaAI/AI-Scientist-v2 to foster the future development of this transformative technology. We also discuss the role of AI in science, including AI safety.

1. Introduction

Automated scientific discovery empowered by artificial intelligence (AI) has garnered considerable attention in recent years (Cornelio et al., 2023; Gil et al., 2014; King et al., 2009; Kitano, 2021; Wang et al., 2023; Xu et al., 2021). The development of end-to-end frameworks capable of autonomously formulating hypotheses, performing experiments, analyzing results, and authoring manuscripts could fundamentally transform the scientific process. A notable recent advance in this direction is The AI Scientist-v1 (Lu et al., 2024), which demonstrated the feasibility of a fully automated scientific workflow and downstream manuscript production. However, significant limitations constrained its broad applicability and autonomy. Specifically, it relied heavily on human-authored code templates requiring manual effort to create a new template for each new topic area. Furthermore, its linear and shallow experimentation approach prevented deeper exploration of scientific hypotheses.

In this paper, we introduce The AI Scientist-v2, a substantially improved successor that directly addresses these limitations. Our contributions are threefold. First, we eliminate the dependency on human-provided code templates, significantly increasing the system's autonomy and ability to be deployed out of the box across multiple machine learning domains. Second, we introduce an experiment manager agent coupled with a novel agentic tree-search algorithm, enabling deeper and

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